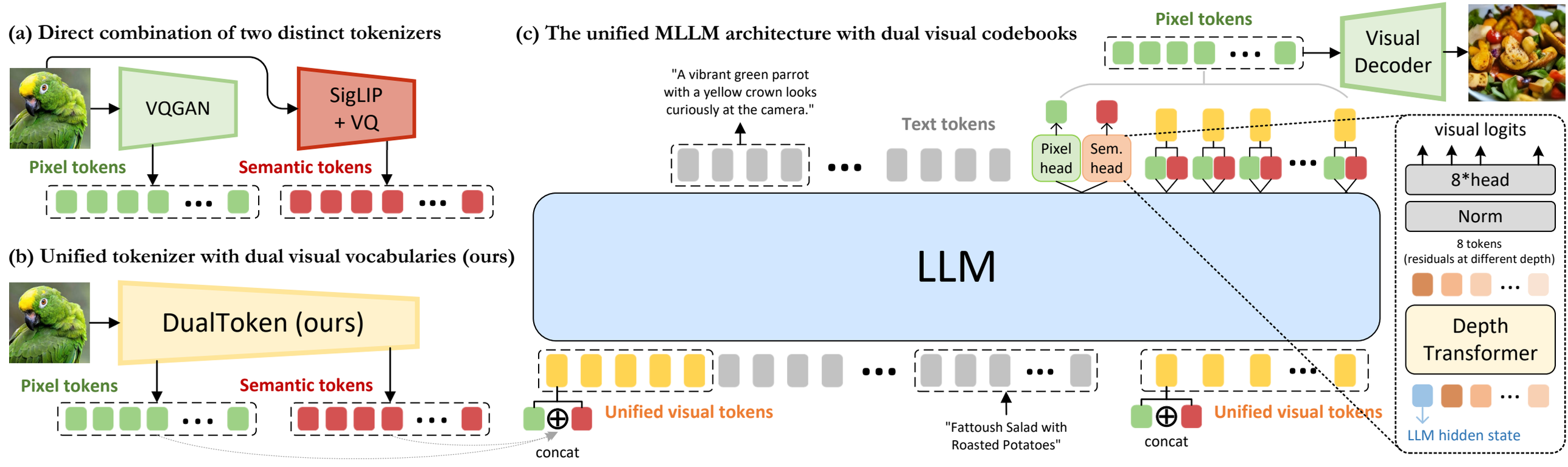
(a) Direct combination of two distinct tokenizers
VQGAN
SigLIP + VQ
Pixel tokens
Semantic tokens
(b) Unified tokenizer with dual visual vocabularies (ours)
DualToken (ours)
Pixel tokens
Semantic tokens
concat
(c) The unified MLLM architecture with dual visual codebooks
Pixel tokens
Visual Decoder
"A vibrant green parrot with a yellow crown looks curiously at the camera."
Text tokens
Pixel head
Sem. head
visual logits
8*head
Norm
8 tokens (residuals at different depth)
Depth Transformer
LLM hidden state
LLM
Unified visual tokens
"Fattoush Salad with Roasted Potatoes"
concat
Unified visual tokens